Work update

Co-electrosynthesis of hydroxylamine and FDCA in a bipolar membrane electrode assembly

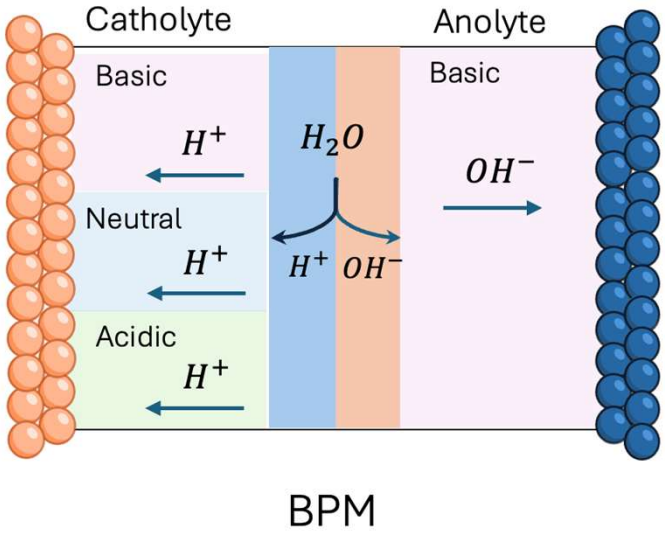
By:
Lorenzo, Tracy, and Hamed

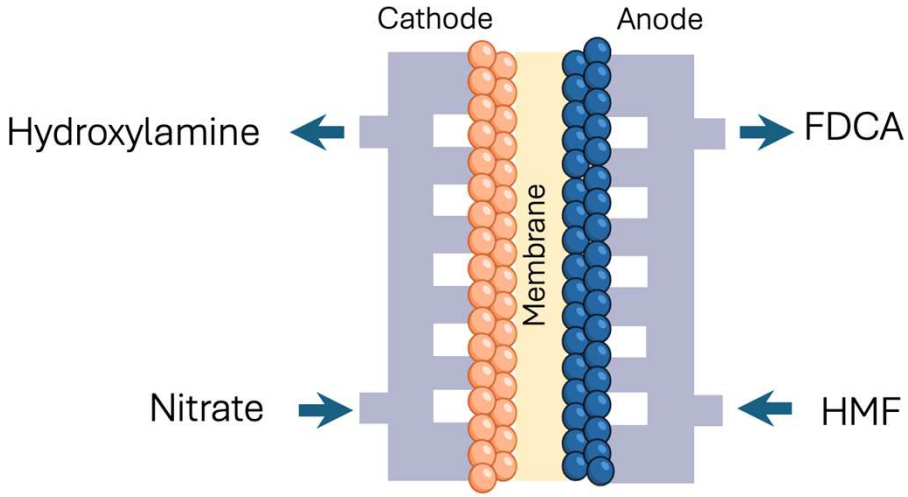
Supervisor:
Prof. Ali Seifitokaldani



July 16, 2026

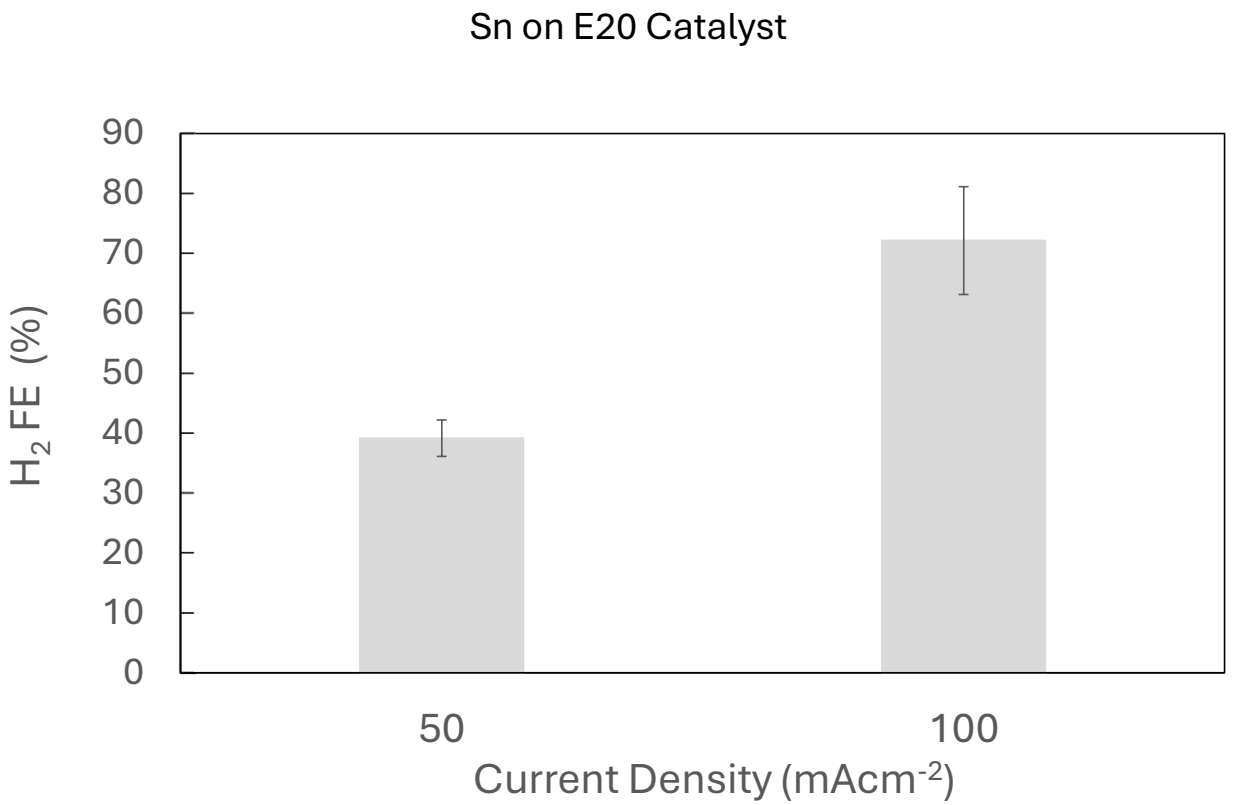
Project concept





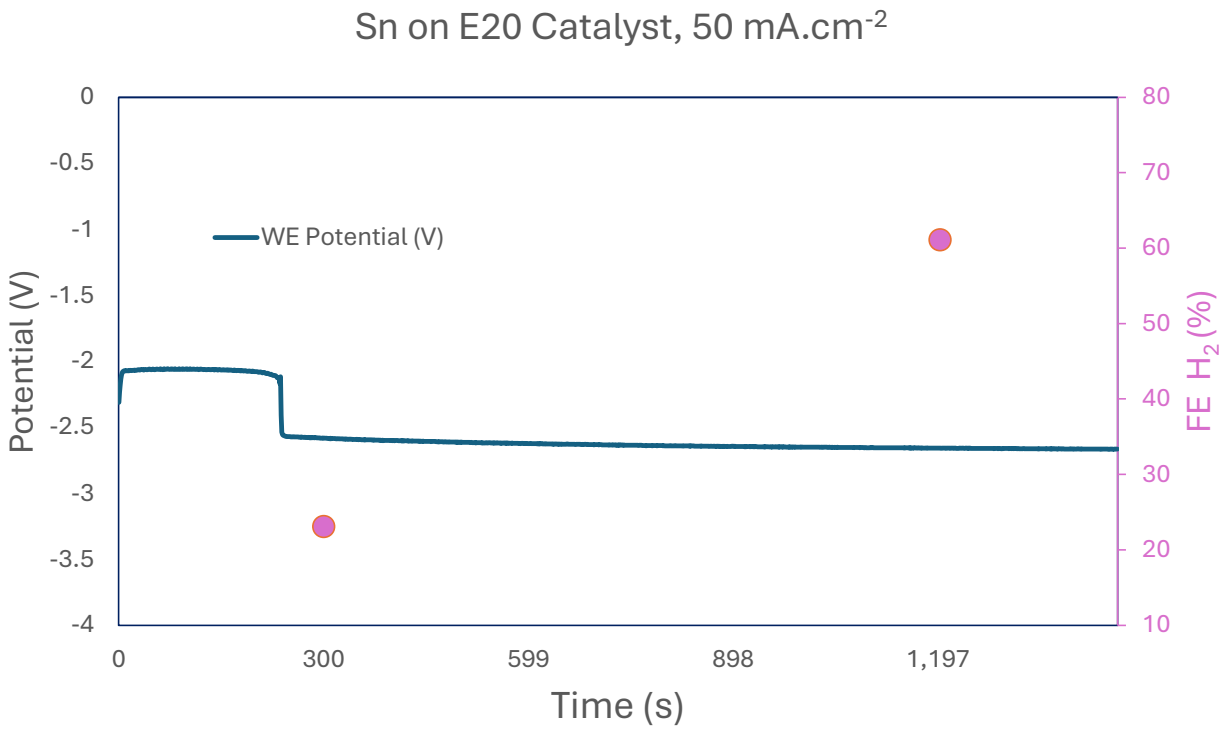
Results – Nitrate Production Performance

Catholyte: 0.1 M KNO_3 + 0.05 M H_2SO_4
Cathode: Sn on E20
Flow rate: 40 mL/min
Anolyte: KOH
Anode: Ni Foam



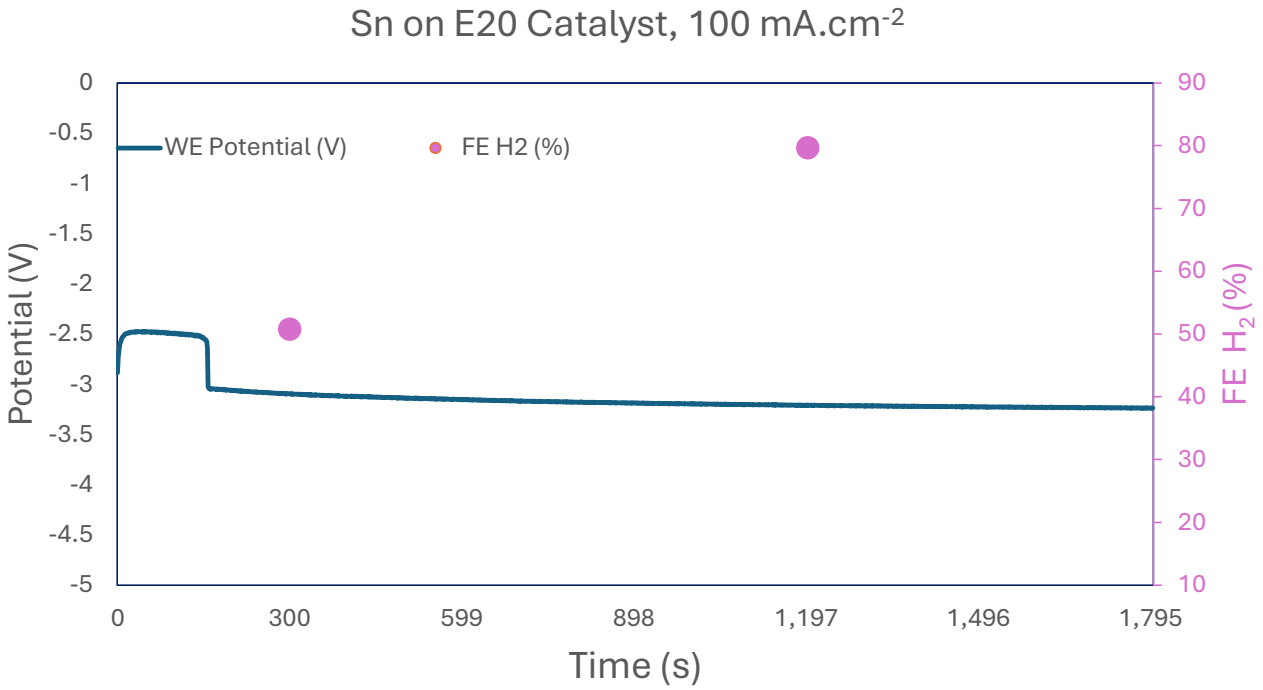
Results – Performance over time

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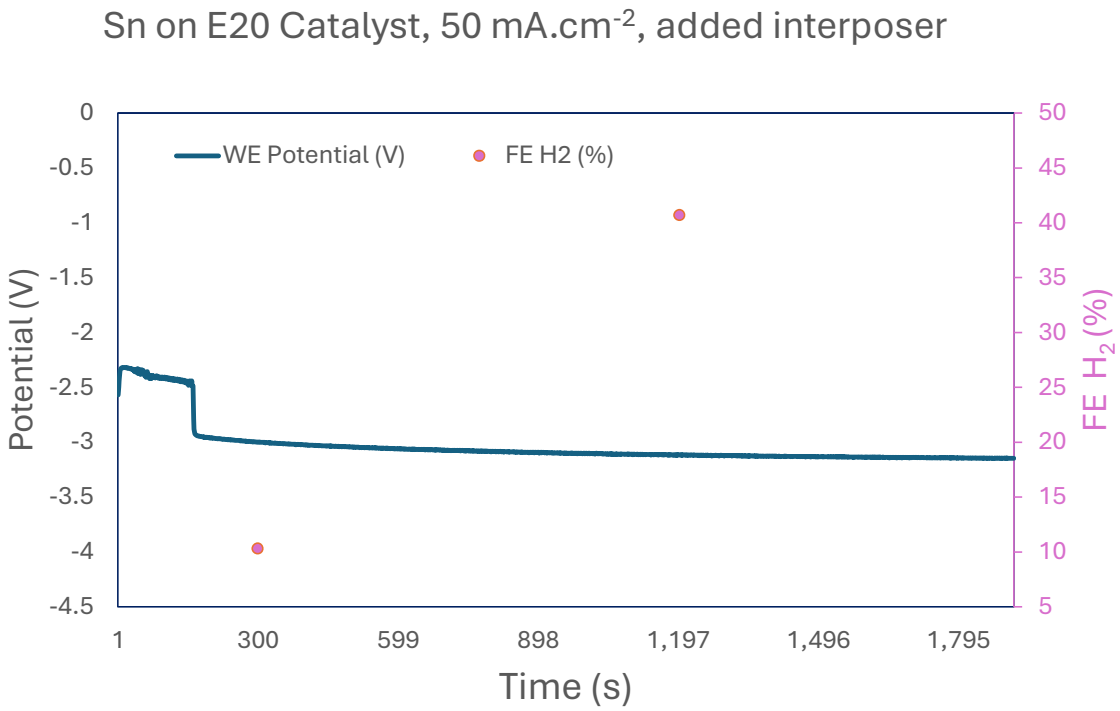
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Interposer



NO3RR Summary 1

Sn on E20 does not perform well with hydrogen production dominating the product selectivity. This is likely due to catalyst decomposition which can be seen as a decrease in current at ~300 seconds.

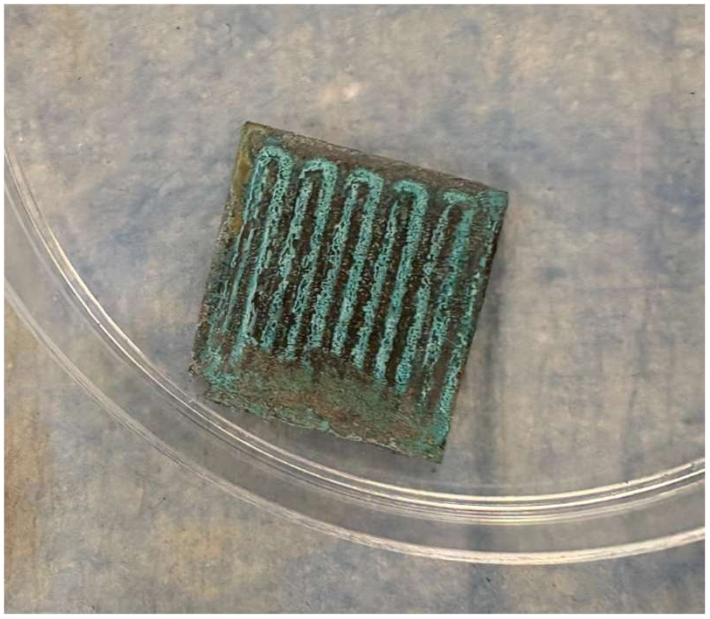
Titanium is another option:

- Studies show that it is a good catalyst for NH_2OH
- We can use a mesh version for increased mass transport
- Supresses hydrogen production

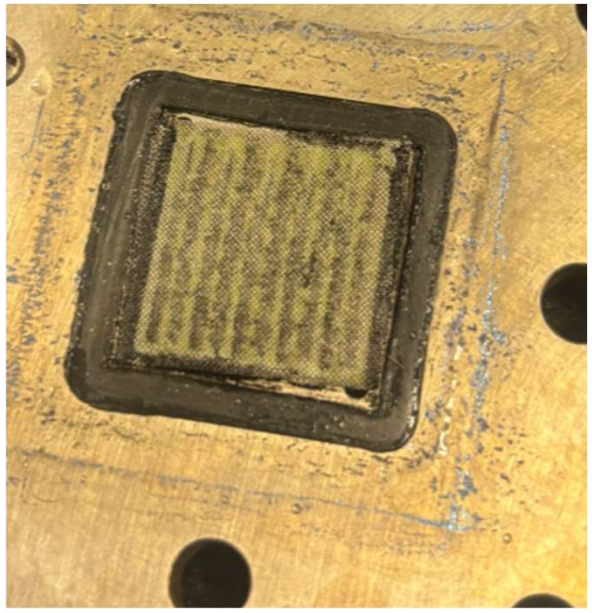
HMF oxidation

Ran several experiments and noted electrolyte color and catalyst degradation each time but we have yet to analyze the liquid results.
Anolyte: 0.1 M HMF + 0.4 M K_2SO_4, Catholyte: 0.1 M KNO_3

NiP on electrochemically deposited onto Ni Foam – 100 $mA.cm^{-2}$



Nickel Mesh - 100 $mA.cm^{-2}$



HMF Oxidation Summary 2

The nickel mesh performed better than the nickel foam, likely due to improved mass transport.

We need to improve the stability of the catalyst:

- Adding an interposer to reduce catalyst contact
- High temperature treatment
- Galvanic replacement with Pd – improve catalyst stability and lowers operating potential

https://www.nature.com/articles/s41467-024-50325-w

Future steps

NO_3RR to hydroxylamine:

Sn on E20 does not seem to be a good catalyst

Future steps: use a titanium mesh

HMF oxidation:

Ni mesh performed much better than Ni foam.

Future steps: Efficiency could be increased by adding an interposer and palladium to the nickel mesh to prevent catalyst degradation, also high temperature treatment of catalyst in oven.

Liquid results still need to be analyzed.